Evergreen Project

Seed Exchange Inventory

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01 Program Information

What is our project actually about?

Credit: LA Times

Our Beneficiary: Lopez Urban Farm

Our beneficiary, Lopez Urban Farm is a non-profit farm 3 miles east of Cal Poly Pomona. They plant crops for the public which allow those in need to harvest at no cost. It crowdfund through donations.

Issues at Lopez Urban Farms

Source: Our phone

When we visited the farm, we noticed the seed exchange program was lacking a proper inventory management system and also lacking an information sheet that contained instructions for how to properly care for each seed.

Our Solution

After feedback from the Operations Manager, Steven, we came to the solution of creating a seed inventory log, that would save information from users. This would allow the farm's workers to quickly track inventory.

Source: iStock

02 Contributions

How does our program help the farm?

Impact

Our program will allow contributors to monitor inventory of each seed by logging when they are adding or removing seeds and it will provide crucial information to the contributor on how to properly care for and maintain each seed.

Source: Gardening know how

03 Details/Design

How did we implmement our program?

What does our program do?

Scan a QR Code

Each seed box has a unique QR code attached. Users can scan it with their phoen to instantly acces that specific seed’s page.

Updates Inventory in Real Time

Users can take a seed package (-1) or add a seed package (+1). This sends a request to our backend, which updates the database.

View Seed Care Information

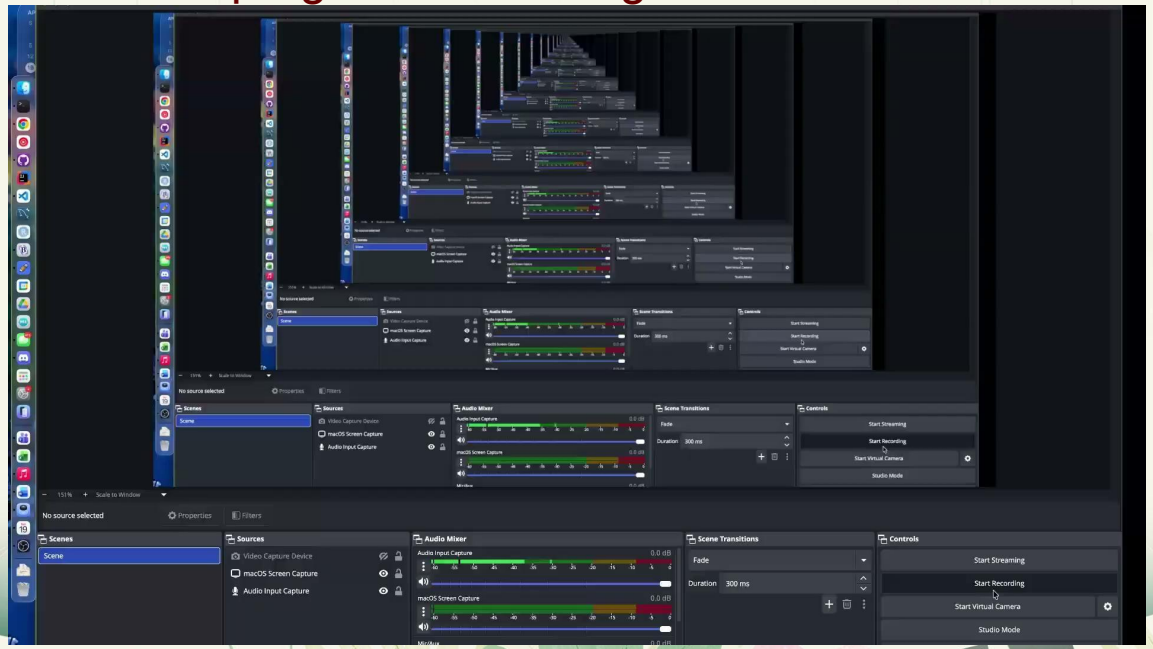
Each box displays a plant description, watering and sunlight needs, and the growth timeline. This greatly simplifies the process for users that are planting seeds, and should provide a easy starting point for those growing seeds.

Centralized System for Workers

Workers can monitor all 27 boxes from one dashboard on their phone, and see which seeds are low or out of stock. They can also update seed types and notes. This prevents them from manually checking every box.

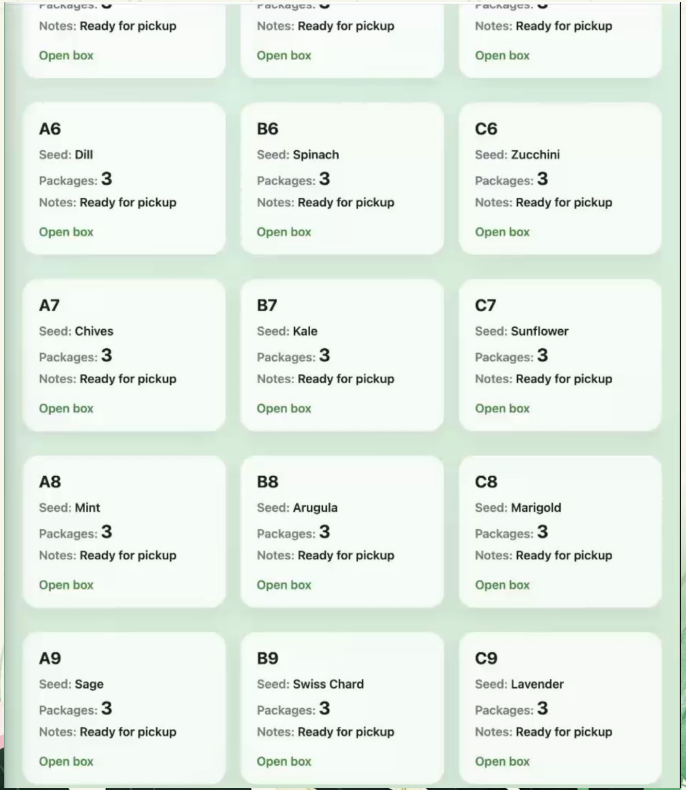
Implementation and Demo

Demo our program here using local host.



Program Interface

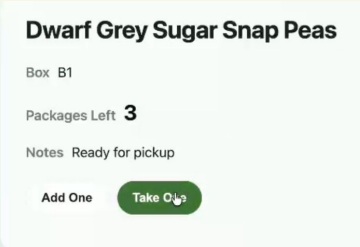
Our interface provides a table containing each seed type, seed type availability and care instructions for each seed.

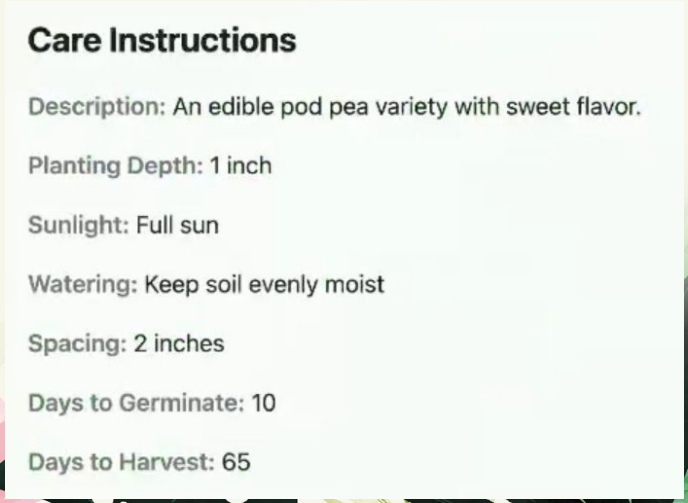


Seed Directory Inventory Logging and Care Instructions

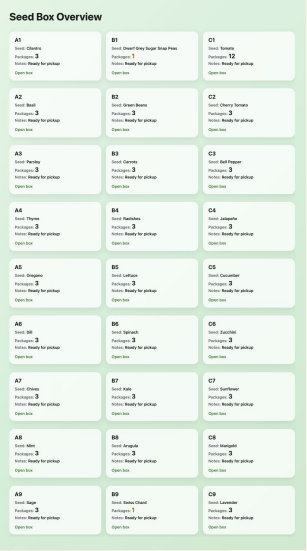
The care instructions are automatically listed below the inventory management.

Users can add and remove seeds.





Data Structure



List of Objects

We represented each seed box as an object, replicating the actual 3x9 seed exchange structure.

Each object groups all of its data, like label, location, and inventory.

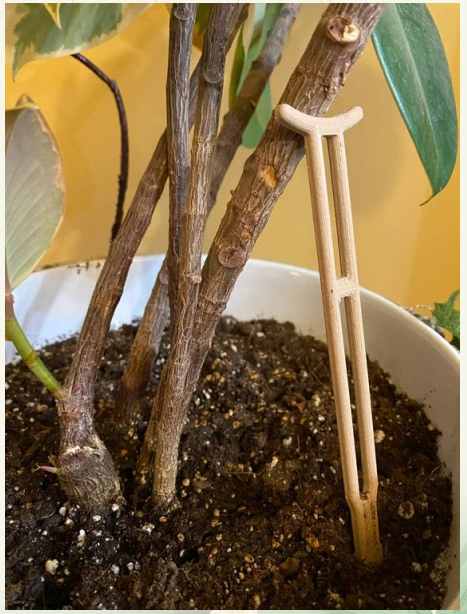
Our backend holds each of these objects in a list, allowing us to easily retrieve, display, and update all 27 boxes.

04 Improvements

How can our program be better?

Areas of Improvement

- Design a more visually appealing interface, maybe even an app
- Might be a way to implement AI interface, so that the user can ask questions about the plants directly.
- Email the client when stock is out.
- Authentication for staff. Our current program uses a shared password, which is Level 0 and extremely vulnerable. Ideally, staff would log in through two-factor authentication.



Source: Etsy

05 Setbacks

What setbacks did we face?

Challenges Faced

Our biggest challenge was time. We started this project late, and found that we lacked time after midterms. Also, it was difficult to get in touch with Steven at first, as well as identifying a problem that actually worked. Steven also seemed slightly hesitant to “marry the dead with the living” in his words.

Source: Dreamstime

Solutions

We resolved our difficulties by putting in a lot of time into the project over a long weekend. Once we did manage to get in contact with Steven, he proved to be an enthusiastic communicator. Also our small group sized allowed us to communicate quickly.

Source: Ourstate

06 Scalability

What does the future of the project look like?

Next Steps

Expand beyond 27 boxes

We could support multiple seed stations, with their own grid of boxes.

Multi-Farm / Commercial Use

Many different farms could use the same system for their seed exchanges. Our app would also work for managing seed inventory for commercial farms.

Mobile Friendly Interface

Our UI is slightly clunky on phones, and could be better optimized since most users will be scanning on their phones.

Staff Authentication

As mentioned earlier, we only have a shared password for staff to edit the seed boxes. This is risky, having a login system with two-factor authentication would be much more secure.

Source: Little Passports

Conclusion

Problem

Lopez Urban Farm’s seed exchange program suffered from a lack of inventory management. Workers and customers would have to check each box manually to check inventory.

Solution

Our inventory tracking program uses lists to manage each seed box as an object to store inventory information. This allows workers to quickly check inventory, as well as allowing customers to check whether a seed is in stock or not. The customers would also gain quick and easy access to seed care by simply scanning a QR code.

Thank You!